

Prop fn exists var aleatoria

Proposición 1. Sea $F: \mathbb{R} \rightarrow \mathbb{R}$ monótona no decreciente que cumple (1), (2) y (3)

$$\implies \exists (\Omega, \Sigma, \mathbb{P}) \text{ espacio de probabilidad} : \exists X \text{ v.a.} : \forall t \in \mathbb{R} : F_X(t) = F(t).$$

Demostración: Consideramos en el espacio de probabilidad $([0, 1], \mathcal{B}([0, 1]), m)$ la variable aleatoria $X: [0, 1] \rightarrow \mathbb{R}$ dada por $\forall \omega \in [0, 1] : X(\omega) = \sup \{t \in \mathbb{R} : F(t) < \omega\}$.

$$\begin{aligned} \implies \forall t \in \mathbb{R} : m(\{\omega \in \mathbb{R} : X(\omega) \leq t\}) &= m(\{\omega \in \mathbb{R} : \sup \{s \in \mathbb{R} : F(s) < \omega\} \leq t\}) \\ &= m(\{\omega \in \mathbb{R} : \omega \leq F(t)\}) = F(t). \end{aligned}$$

Por tanto, $\forall t \in \mathbb{R} : F_X(t) = m(\{\omega \in \mathbb{R} : X(\omega) \leq t\}) = F(t)$ ¹. ■

¹También se puede definiendo X como la identidad en el espacio de medida dado por la medida de Lebesgue-Stieltjes.

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